

1. Product and Company Identification

Product identifier	Methyl Alcohol
Version #	01
Issue date	06-09-2014
Chemical name	Methyl Alcohol
Chemical description	Saturated primary aliphatic alcohol
MSDS Number	COM087
Product use	Professional use only
Synonym(s) Manufacturer	Carbinol * Wood naphtha * METHANOL
information Supplier	Refer to supplier Comet Chemical 3463 Thomas Street Innisfill, ON L9S 3W4 CA Information (M-F 8:00-5:00): 705-436-5580 24 Hour Number (Newalta): 800-567-7455

2. Hazards Identification

Emergency overview	DANGER Flammable liquid and vapor. Vapors may cause flash fires. POISON. May be fatal or cause blindness if swallowed. Cannot be made nonpoisonous. Harmful by inhalation and in contact with skin. May cause central nervous system effects. Causes eye irritation. Prolonged or repeated overexposure may cause liver effects. Teratogenic.
Potential health effects	
Routes of exposure	Inhalation. Ingestion. Skin contact. Eye contact.
Eyes	Causes eye irritation.
Skin	Harmful in contact with skin. May cause mild skin irritation. Skin absorption: May be absorbed and cause symptoms similar to those for inhalation.
Inhalation	May be harmful if inhaled. Toxic effects exerted upon nervous system, particularly the optic nerve. May cause irritation of respiratory tract.
Ingestion	Toxic if swallowed. Even small amounts (30-250 ml methanol) may be fatal. Symptoms are stomach ache, nausea, vomiting, dullness, visual disorder and blindness.
Target organs	Central nervous system. Eyes. Gastro-intestinal tract. Respiratory system. Skin.
Chronic effects	Pregnant women or women of child-bearing age should not be exposed to this product. May cause birth defects. Frequent or prolonged contact may defat and dry the skin, leading to discomfort and dermatitis. Prolonged or repeated overexposure may cause liver effects.
Signs and symptoms	Causes moderate eye irritation. Symptoms may include stinging, tearing, redness, swelling, and blurred vision. May cause mild skin irritation. Symptoms may include redness, edema, drying, defatting and cracking of the skin. May cause respiratory irritation. Symptoms may include pain, headache, nausea, vomiting, dizziness, drowsiness and other central nervous system effects. May be fatal or cause blindness if swallowed. Cannot be made nonpoisonous. Ingestion of large amounts may cause nausea, vomiting, diarrhea, as well as depression of the central nervous system.
Potential environmental effects	May cause long-term adverse effects in the environment.

3. Composition / Information on Ingredients

Components	CAS #	Percent
Methyl Alcohol	67-56-1	100

4. First Aid Measures

First aid procedures	
Eye contact	Immediately flush eyes with plenty of water for at least 20 minutes. Seek immediate medical attention/advice.

Skin contact	Immediately flush with plenty of water for at least 15 minutes while removing contaminated clothing and shoes. Get medical attention. Wash contaminated clothing before reuse.
Inhalation	Move to fresh air. Oxygen or artificial respiration if needed. Get medical attention, if needed.
Ingestion	IF SWALLOWED: Immediately call a POISON CENTER or doctor/physician. Do not induce vomiting. Never give anything by mouth to a victim who is unconscious or is having convulsions.
Notes to physician	Treat symptomatically. Immediate medical attention is required. This product is a CNS depressant. Contains methanol. Onset of symptoms may be delayed for 18 to 24 hours after ingestion. Medical supervision for minimum 48 hours. Administration of ethanol can slow the metabolism of methanol, thus reducing the potential for harmful effects.
General advice	If you feel unwell, seek medical advice (show the label where possible). Ensure that medical personnel are aware of the material(s) involved, and take precautions to protect themselves. Show this safety data sheet to the doctor in attendance.

5. Fire Fighting Measures

Flammable properties	Flammable by WHMIS criteria. Heat may cause the containers to explode. Vapors are heavier than air and may spread along floors. Vapors may travel considerable distance to a source of ignition and flash back.
Extinguishing media	
Suitable extinguishing media	Alcohol resistant foam. Water fog. Dry chemical powder. Carbon dioxide (CO ₂).
Unsuitable extinguishing media	Do not use water jet as an extinguisher, as this will spread the fire.
Protection of firefighters	
Specific hazards arising from the chemical	Fire may produce irritating, corrosive and/or toxic gases.
Protective equipment for firefighters	Firefighters should wear full protective clothing including self contained breathing apparatus.
Fire fighting equipment/instructions	In case of fire and/or explosion do not breathe fumes. Firefighters must use standard protective equipment including flame retardant coat, helmet with face shield, gloves, rubber boots, and in enclosed spaces, SCBA. Ventilate the contaminated area. Evacuate area and fight fire from a safe distance. Cool containers exposed to heat with water spray and remove container, if no risk is involved. Prevent fire extinguishing water from contaminating surface water or the ground water system.
Specific methods	Cool containers exposed to flames with water until well after the fire is out.
Explosion data	
Sensitivity to static discharge	May be sensitive to static discharge. Vapours in the flammable range may be ignited by a static discharge or sufficient energy.
Sensitivity to mechanical impact	Not expected to be sensitive to mechanical impact.
Hazardous combustion products	Carbon oxides. Formaldehyde. Other unidentified organic compounds.

6. Accidental Release Measures

Personal precautions	Keep people away from and upwind of spill/leak. Consider initial downwind evacuation for at least 500 meters (1/3 mile). Keep unnecessary personnel away. Eliminate all ignition sources (no smoking, flares, sparks, or flames in immediate area). Wear appropriate protective equipment and clothing during clean-up. Do not breathe mist or vapor. Ventilate closed spaces before entering them. Use appropriate containment to avoid environmental contamination. Local authorities should be advised if significant spillages cannot be contained. For personal protection, see section 8 of the MSDS.
Environmental precautions	Prevent further leakage or spillage if safe to do so. Do not contaminate water.
Methods for containment	Eliminate all ignition sources (no smoking, flares, sparks, or flames in immediate area). Stop leak if you can do so without risk. Move the cylinder to a safe and open area if the leak is irreparable. Dike the spilled material, where this is possible. Prevent entry into waterways, sewer, basements or confined areas.

Methods for cleaning up

Extinguish all flames in the vicinity. Should not be released into the environment. This product is miscible in water.

Large Spills: Stop the flow of material, if this is without risk. Dike the spilled material, where this is possible. Cover with plastic sheet to prevent spreading. Use a non-combustible material like vermiculite, sand or earth to soak up the product and place into a container for later disposal. Following product recovery, flush area with water.

Small Spills: Absorb residues with a non-combustible absorbent material (e.g. sand, vermiculite) and collect adsorbate for disposal. Clean surface thoroughly to remove residual contamination.

Never return spills to original containers for re-use. Clean up in accordance with all applicable regulations. For waste disposal, see section 13 of the MSDS.

Other information

Clean up in accordance with all applicable regulations.

7. Handling and Storage

Handling

Obtain special instructions before use. Do not handle until all safety precautions have been read and understood. Do not handle, store or open near an open flame, sources of heat or sources of ignition. Protect material from direct sunlight. All equipment used when handling the product must be grounded. Take precautionary measures against static discharges. Handling operations that can promote accumulation of static charges include but are not limited to: mixing, filtering, pumping at high flow rates, splash filling, creating mists or sprays, tank and container filling, tank cleaning, sampling, gauging, switch loading, vacuum truck operations. Use non-sparking tools and explosion-proof equipment. Do not breathe mist or vapor. Avoid contact with eyes, skin, and clothing. Do not taste or swallow. When using, do not eat, drink or smoke. Use only in area provided with appropriate exhaust ventilation. Wear appropriate personal protective equipment. Observe good industrial hygiene practices. Wash contaminated clothing before reuse.

Storage

Store locked up. Keep away from heat, sparks and open flame. Prevent electrostatic charge build-up by using common bonding and grounding techniques. Eliminate sources of ignition. Ground/bond container and equipment. These alone may be insufficient to remove static electricity. Store in original tightly closed container. Store in a cool, dry place out of direct sunlight. Store in a well-ventilated place. Store away from incompatible materials (see Section 10 of the MSDS). Keep in an area equipped with sprinklers.

8. Exposure Controls / Personal Protection

Occupational exposure limits

US. ACGIH Threshold Limit Values

Material	Type	Value
Methyl Alcohol	STEL	250 ppm
	TWA	200 ppm

Components	Type	Value
Methyl Alcohol (CAS 67-56-1)	STEL	250 ppm
	TWA	200 ppm

US. OSHA Table Z-1 Limits for Air Contaminants (29 CFR 1910.1000)

Material	Type	Value
Methyl Alcohol	PEL	260 mg/m3 200 ppm
Components	Type	Value
Methyl Alcohol (CAS 67-56-1)	PEL	260 mg/m3 200 ppm

Biological limit values

ACGIH Biological Exposure Indices

Material	Value	Determinant	Specimen	Sampling Time
Methyl Alcohol	15 mg/l	Methanol	Urine	*
Components	Value	Determinant	Specimen	Sampling Time
Methyl Alcohol (CAS 67-56-1)	15 mg/l	Methanol	Urine	*

* - For sampling details, please see the source document.

Exposure guidelines

Canada - Alberta OELs: Skin designation

Methyl Alcohol (CAS 67-56-1)

Can be absorbed through the skin.

Canada - British Columbia OELs: Skin designation

Methyl Alcohol (CAS 67-56-1)

Can be absorbed through the skin.

Canada - Manitoba OELs: Skin designation

Methyl Alcohol (CAS 67-56-1)

Can be absorbed through the skin.

Canada - Ontario OELs: Skin designation

Methyl Alcohol (CAS 67-56-1)

Can be absorbed through the skin.

Canada - Quebec OELs: Skin designation

Methyl Alcohol (CAS 67-56-1)

Can be absorbed through the skin.

Canada - Saskatchewan OELs: Skin designation

Methyl Alcohol (CAS 67-56-1)

Can be absorbed through the skin.

US ACGIH Threshold Limit Values: Skin designation

Methyl Alcohol (CAS 67-56-1)

Can be absorbed through the skin.

Engineering controls

Good general ventilation (typically 10 air changes per hour) should be used. Ventilation rates should be matched to conditions. If applicable, use process enclosures, local exhaust ventilation, or other engineering controls to maintain airborne levels below recommended exposure limits. If exposure limits have not been established, maintain airborne levels to an acceptable level. Ensure adequate ventilation, especially in confined areas.

Personal protective equipment

Eye/face protection

Chemical goggles are recommended. A full face shield may also be necessary. Eye wash facilities and emergency shower must be available when handling this product.

Skin protection

Wear appropriate chemical resistant clothing. Use of an impervious apron is recommended.

Respiratory protection

Use a NIOSH/MSHA approved respirator if there is a risk of exposure to dust/fume at levels exceeding the exposure limits. Advice should be sought from respiratory protection specialists.

Hand protection

Wear appropriate chemical resistant gloves. Advice should be sought from glove suppliers.

9. Physical & Chemical Properties

Appearance

Clear, colorless liquid with alcohol-like odor.

Physical state

Liquid.

Form

Liquid.

Color

Clear colorless or nearly colorless

Odor

Alcohol-like.

Odor threshold

50 - 100 ppm

pH

Not available.

Vapor pressure

92 mm Hg at 20 °C

Vapor density

> 1.1

Boiling point

148.46 °F (64.7 °C) 101.325 kPa

Melting point/Freezing point

-144.04 °F (-97.8 °C)

Solubility (water)

Soluble

Specific gravity

0.79 at 20 °C

Relative density

Not available.

Flash point

53.6 °F (12.0 °C) Closed Cup

Flammability limits in air, upper, % by volume

36

Flammability limits in air, lower, % by volume

7.3

Auto-ignition temperature

867.2 °F (464 °C)

Evaporation rate

< 1

Partition coefficient (n-octanol/water)

-0.8

Molecular weight

32.04 g/mol

Molecular formula

C-H4-O

Other data

Density	0.79 g/cm ³
Dynamic viscosity	0.61 mPa.s
Kinematic viscosity	0.7807 mm ² /s estimated
Solubility (other)	Soluble in all proportions in ethanol, other alcohols, benzene, chloroform, diethyl ether, other ethers, esters, ketones, and most other organic solvents.

10. Chemical Stability & Reactivity Information

Chemical stability	Material is stable under normal conditions. Risk of explosion.
Conditions to avoid	Avoid heat, sparks, open flames and other ignition sources. Avoid temperatures exceeding the flash point. Contact with incompatible materials. Do not use in areas without adequate ventilation.
Incompatible materials	Strong oxidizing agents. Acids. Powdered metal. Alkali metals. Hydrogen peroxide. Isocyanates. Mineral acids. Potassium persulfate. Perchloric Acid.
Hazardous decomposition products	None known, refer to hazardous combustion products in Section 5. In the event of fire the following can be released: Carbon monoxide. Formaldehyde.
Possibility of hazardous reactions	Hazardous polymerization does not occur.

11. Toxicological Information**Toxicological data**

Components	Species	Test Results
Methyl Alcohol (CAS 67-56-1)		
Acute		
<i>Dermal</i>		
LD50	Monkey	> 393 mg/kg
	Rabbit	15800 mg/kg
<i>Inhalation</i>		
LC50	Rat	> 4.1 mg/l/4h (vapor)
<i>Oral</i>		
LD50	Human	300 - 1000 mg/kg (estimated human lethal dose)
	Rat	5628 mg/kg
Acute effects	May cause moderate eye irritation. May cause mild skin irritation. May cause respiratory irritation. May cause irritation of the gastrointestinal tract. May be fatal or cause blindness if swallowed.	
Sensitization	Not expected to be a skin or respiratory sensitizer.	
Chronic effects	Frequent or prolonged contact may defat and dry the skin, leading to discomfort and dermatitis. Prolonged or repeated overexposure may cause liver effects.	
Carcinogenicity	This product is not considered to be a carcinogen by IARC, ACGIH, NTP, or OSHA.	
Skin corrosion/irritation	May be irritating to the skin.	
Serious eye damage/irritation	May cause moderate eye irritation.	
Mutagenicity	No data available to indicate product or any components present at greater than 0.1% are mutagenic or genotoxic.	
Reproductive effects	Contains methanol, which may cause teratogenic effects at doses which are not maternally toxic. Effects were observed following inhalation of high concentrations of methanol, as doses which were not maternally toxic.	
Teratogenicity	Hazardous by WHMIS criteria. Avoid exposure to women during early pregnancy.	
Synergistic materials	Not available.	
Further information	Symptoms may be delayed.	

12. Ecological Information

Ecotoxicological data

Components	Species		Test Results
Methyl Alcohol (CAS 67-56-1)			
Aquatic			
Crustacea	EC50	Water flea (Daphnia magna)	> 10000 mg/l, 48 hours
<i>Acute</i>			
Algae	EC50	Green algae (Scenedesmus quadricauda)	> 1000 mg/l, 96 hours
Fish	LC50	Fathead minnow (Pimephales promelas)	15400 mg/l, 96 hours
Ecotoxicity	Not expected to cause hazardous effects to the environment.		
Environmental effects	An environmental hazard cannot be excluded in the event of unprofessional handling or disposal.		
Aquatic toxicity	The product is not classified as environmentally hazardous. However, this does not exclude the possibility that large or frequent spills can have a harmful or damaging effect on the environment. The product should not be allowed to enter drains, water courses or the soil.		
Persistence and degradability	Readily biodegradable.		
Partition coefficient			
Methyl Alcohol		-0.77	
Methyl Alcohol		-0.77	
Mobility in environmental media	High water solubility indicates a high mobility in soil.		

13. Disposal Considerations

Disposal instructions	Collect and reclaim or dispose in sealed containers at licensed waste disposal site. Do not allow this material to drain into sewers/water supplies. Dispose in accordance with all applicable regulations.
Waste from residues / unused products	Dispose of in accordance with local regulations. Empty containers or liners may retain some product residues. This material and its container must be disposed of in a safe manner (see: Disposal instructions).
Contaminated packaging	Empty containers should be taken to an approved waste handling site for recycling or disposal. Since emptied containers may retain product residue, follow label warnings even after container is emptied.

14. Transport Information

TDG	
UN number	UN1230
UN proper shipping name	METHANOL
Transport hazard class(es)	
Class	3
Subsidiary risk	6.1(PGI, II)
Packing group	II
Environmental hazards	Not available.
Special precautions for user	Read safety instructions, MSDS and emergency procedures before handling.
For shipments by ground within Canada, the Limited Quantity / Consumer Commodity Exemption may apply for containers which hold 5 Litres or less of the material. Refer to TDG Section 1.17 for details.	
IATA	
UN number	UN1230
UN proper shipping name	Methanol
Transport hazard class(es)	
Class	3
Subsidiary risk	6.1(PGI, II)
Packing group	II
Environmental hazards	No.
ERG Code	3L
Special precautions for user	Read safety instructions, MSDS and emergency procedures before handling.
Other information	
Passenger and cargo aircraft	Allowed.

Cargo aircraft only

Allowed.

IMDG

UN number	UN1230
UN proper shipping name	METHANOL
Transport hazard class(es)	
Class	3
Subsidiary risk	6.1(PGI, II)
Packing group	II
Environmental hazards	
Marine pollutant	No.
EmS	F-E, S-D
Special precautions for user	Read safety instructions, MSDS and emergency procedures before handling.

IATA; IMDG; TDG



15. Regulatory Information

Canadian regulations

This product has been classified in accordance with the hazard criteria of the CPR and the MSDS contains all the information required by the CPR.

WHMIS status

Controlled

WHMIS classification

B2 - Flammable Liquids
D1B - Immediate/Serious-TOXIC
D2A - Other Toxic Effects-VERY TOXIC
D2B - Other Toxic Effects-TOXIC

WHMIS labeling



International Inventories

Country(s) or region	Inventory name	On inventory (yes/no)*
Australia	Australian Inventory of Chemical Substances (AICS)	Yes
Canada	Domestic Substances List (DSL)	Yes
Canada	Non-Domestic Substances List (NDSL)	No
China	Inventory of Existing Chemical Substances in China (IECSC)	Yes
Europe	European Inventory of Existing Commercial Chemical Substances (EINECS)	Yes
Europe	European List of Notified Chemical Substances (ELINCS)	No
Japan	Inventory of Existing and New Chemical Substances (ENCS)	Yes
Korea	Existing Chemicals List (ECL)	Yes
New Zealand	New Zealand Inventory	Yes
Philippines	Philippine Inventory of Chemicals and Chemical Substances (PICCS)	Yes
United States & Puerto Rico	Toxic Substances Control Act (TSCA) Inventory	Yes

*A "Yes" indicates that all components of this product comply with the inventory requirements administered by the governing country(s)

A "No" indicates that one or more components of the product are not listed or exempt from listing on the inventory administered by the governing country(s).

16. Other Information

HMIS® ratings

Health: 2*
Flammability: 3
Physical hazard: 0

NFPA ratings

Health: 1
Flammability: 3
Instability: 0

Disclaimer

Prepared by: ICC The Compliance Center Inc. 1-888-442-9628
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Disclaimer

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Legend to abbreviations and acronyms used in the SDS

ACGIH: American Conference of Governmental Industrial Hygienists
CAS: Chemical Abstract Services
CEPA: Canadian Environmental Protection Act
CPR: Controlled Products Regulation
CSA: Canadian Standards Association
DSL: Domestic Substance List
HMIS: Hazardous Materials Identification System
HPA: Hazardous Protection Act
HSDB: Hazardous Substance Data Bank
IARC: International Agency for Research on Cancer
IATA: International Air Transport Association
ICAO: International Civil Aviation Organisation
IMDG: International Maritime Dangerous Goods
LC: Lethal Concentration
LD: Lethal Dose
MSDS: Material Safety Data Sheet
NFPA: National Fire Protection Association
NOEC: No observable effect concentration
NTP: National Toxicology Program
OECD: Organisation for Economic Co operation and Development
OEL: National occupational exposure limits
OSHA: Occupational Safety and Health Administration
PPE: Personal Protective Equipment
RCRA: Resource Conservation and Recovery Act
RTECS: Registry of Toxic Effects of Chemical Substances
STEL: Short Term Exposure Limit
TDG: Canadian Transportation of Dangerous Goods Act & Regulations
TWA: Time Weighted Average
WEL: Workplace Exposure Limit
WHMIS: Workplace Hazardous Materials Identification System

References

Canadian Centre for Occupational Health and Safety, CCInfoWeb Databases, 2014 (Chempendium, RTECs, HSDB, INCHEM)
European Chemicals Agency, Classification Legislation, 2014.
Material Safety Data Sheet from manufacturer.
OECD - The Global Portal to Information on Chemical Substances - eChemPortal, 2014.